

Part I

SUMMARY of the Definition of the Derivatives

Summary

February 21, 2026

Summary

- The *average rate of change*, $AvRCh$, of a function f across an interval $[a, b]$ is given by

$$\begin{aligned} AvRCh &= \frac{\Delta f}{\Delta x} \\ &= \frac{f(b) - f(a)}{b - a} \end{aligned}$$

- The slope, m_{sec} , of the line secant to the graph of a function f at the points $(a, f(a))$ and $(b, f(b))$ is given by

$$\begin{aligned} m_{sec} &= \frac{\Delta y}{\Delta x} \\ &= \frac{f(b) - f(a)}{b - a} \end{aligned}$$

- The *derivative* of the function f at a , denoted $f'(a)$, is given by

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

and represents the instantaneous rate of change of f with respect to x at $x = a$ and also the slope of the line tangent to the graph of $y = f(x)$ at the point $(a, f(a))$.

Part II

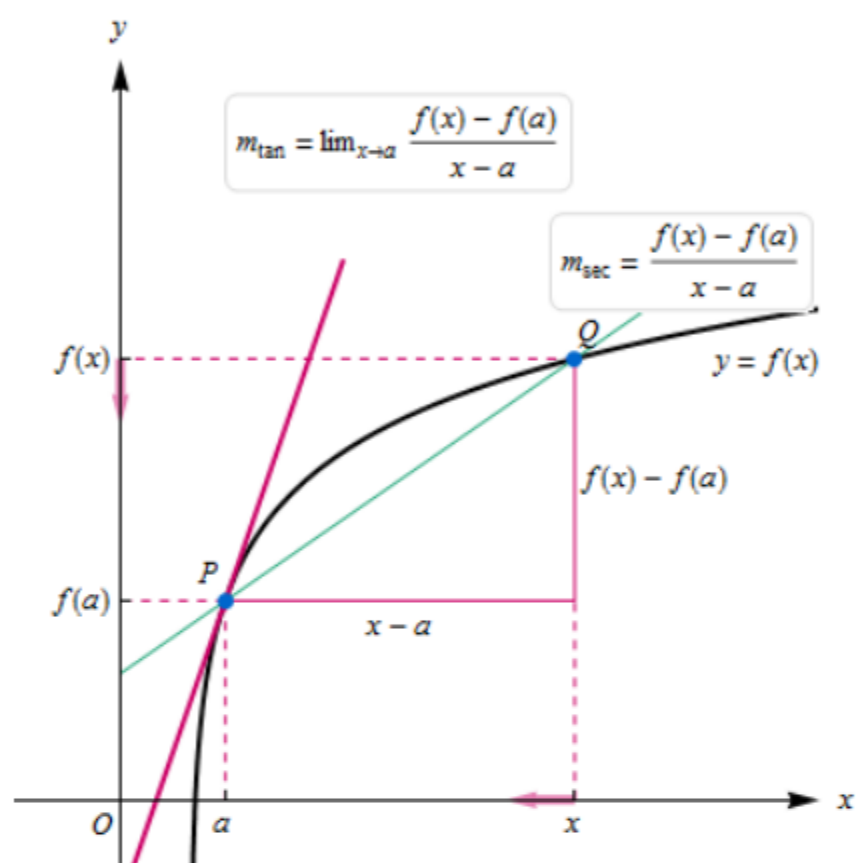
Recitation Questions

Problem 1

February 21, 2026

Problem 1

Problem 1 Consider the following two figures depicting the same graph $y =$



Problem 1

- (a) In Figure 1 on the left, what are the coordinates of P and Q ? In Figure 2 on the right, what are the expressions for the coordinates of P and Q ?

Explanation. Figure 1 on the left: $P(a, f(a))$ and $Q(x, f(x))$

Figure 2 on the right: $P(a, f(a))$ and $Q((a + h), f(a + h))$

- (b) Express the slope of the secant line through P and Q in terms of the above coordinates for each Figure 1 and Figure 2.

Explanation. Figure 1: $\frac{f(x) - f(a)}{x - a}$ Figure 2: $\frac{f(a + h) - f(a)}{h}$

- (c) Express the slope of the tangent line at the point P in terms of the above coordinates for each Figure 1 and Figure 2.

Explanation. Figure 1: $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ Figure 2: $\lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$

- (d) What is the difference?

Explanation. Just the variable we use, x or h where $x = a + h \iff h = x - a$

- (e) For each of the two graphs, which lines are the secant lines?

Explanation. The green line is the secant line in each of the two graphs.

- (f) For each of the two graphs, which lines are the tangent lines?

Explanation. The red line is the tangent line in each of the two graphs.

Problem 2

February 21, 2026

Problem 2 For each of the following functions find an equation of the tangent line at the given point.

(a) $f(x) = -5x^2 + 7x - 9$ at $x = 3$.

Explanation. The slope of tangent line is given by this limit: $f'(3) = \lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$. Notice that this limit has form $\frac{0}{0}$, so it is indeterminate.

$$\begin{aligned} f'(3) &= \lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} = \lim_{x \rightarrow 3} \frac{(-5x^2 + 7x - 9) - (-45 + 21 - 9)}{x - 3} \\ &= \lim_{x \rightarrow 3} \frac{-5x^2 + 7x - 9 + 33}{x - 3} \\ &= \lim_{x \rightarrow 3} \frac{-5x^2 + 7x + 24}{x - 3} \\ &= \lim_{x \rightarrow 3} \frac{(x - 3)(-5x - 8)}{x - 3} \\ &= \lim_{x \rightarrow 3} (-5x - 8) \\ &= -5(3) - 8 = -23. \end{aligned}$$

Point on tangent line: $(3, f(3)) = (3, -33)$.

An equation of tangent line is:

$$y + 33 = -23(x - 3)$$

(b) $g(u) = \sqrt{5u - 4}$ at $u = 3$.

Explanation. The slope of tangent line is given by: $g'(3) = \lim_{h \rightarrow 0} \frac{g(3 + h) - g(3)}{h}$.

Problem 2

Notice that this limit has form $\frac{0}{0}$, so it is indeterminate.

$$\begin{aligned}
 g'(3) &= \lim_{h \rightarrow 0} \frac{g(3+h) - g(3)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{5(3+h) - 4} - \sqrt{11}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sqrt{5(3+h) - 4} - \sqrt{11}}{h} \cdot \frac{\sqrt{5(3+h) - 4} + \sqrt{11}}{\sqrt{5(3+h) - 4} + \sqrt{11}} \\
 &= \lim_{h \rightarrow 0} \frac{5(3+h) - 4 - 11}{h \left(\sqrt{5(3+h) - 4} + \sqrt{11} \right)} \\
 &= \lim_{h \rightarrow 0} \frac{5h + 15 - 15}{h \left(\sqrt{5(3+h) - 4} + \sqrt{11} \right)} \\
 &= \lim_{h \rightarrow 0} \frac{5}{\left(\sqrt{5(3+h) - 4} + \sqrt{11} \right)} \\
 &= \frac{5}{\sqrt{5(3+0) - 4} + \sqrt{11}} \\
 &= \frac{5}{2\sqrt{11}}.
 \end{aligned}$$

Point on tangent line: $(3, g(3)) = (3, \sqrt{11})$.

Equation of tangent line:

$$y - \sqrt{11} = \frac{5}{2\sqrt{11}}(u - 3)$$

(c) $s(z) = \frac{z}{z-5}$ at $z = 3$.

Explanation. The slope of tangent line is given by: $s'(3) = \lim_{z \rightarrow 3} \frac{s(z) - s(3)}{z - 3}$.

Problem 2

Notice that this limit has form $\frac{0}{0}$, so it is indeterminate.

$$\begin{aligned}s'(3) &= \lim_{z \rightarrow 3} \frac{s(z) - s(3)}{z - 3} \\&= \lim_{z \rightarrow 3} \frac{\frac{z}{z-5} - \frac{3}{3-5}}{z - 3} \\&= \lim_{z \rightarrow 3} \frac{\frac{z}{z-5} + \frac{3}{2}}{z - 3} \\&= \lim_{z \rightarrow 3} \frac{\frac{2z}{2(z-5)} + \frac{3(z-5)}{2(z-5)}}{z - 3} \\&= \lim_{z \rightarrow 3} \frac{\frac{2z+3z-15}{2(z-5)}}{z - 3} \\&= \lim_{z \rightarrow 3} \frac{5z - 15}{2(z - 5)} \cdot \frac{1}{z - 3} \\&= \lim_{z \rightarrow 3} \frac{5(z - 3)}{2(z - 5)} \cdot \frac{1}{z - 3} \\&= \lim_{z \rightarrow 3} \frac{5}{2(z - 5)} \\&= \frac{5}{2(3 - 5)} = -\frac{5}{4}.\end{aligned}$$

Point on tangent line: $(3, s(3)) = (3, -3/2)$.

Equation of tangent line:

$$y + \frac{3}{2} = -\frac{5}{4}(z - 3)$$

Problem 3

February 21, 2026

Problem 3

Problem 3 Find an equation of the tangent line at the given point. Then graph the function and the tangent line on the same plot. $f(x) = \sqrt{x+1}$ at $x = 3$

Explanation. The slope of tangent line is given by: $f'(3) = \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h}$.

This limit has form $\frac{0}{0}$, so it is an indeterminate form.

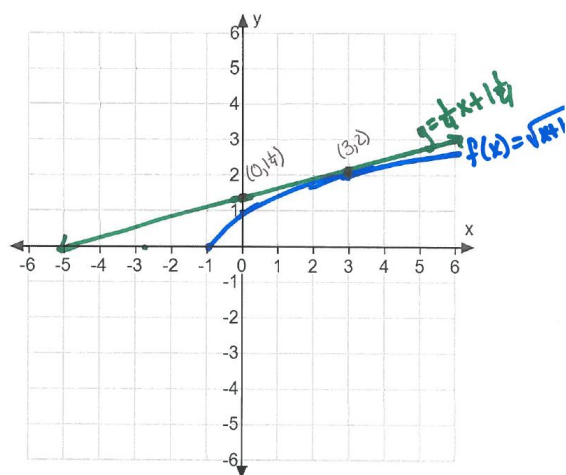
$$\begin{aligned} f'(3) &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{3+h+1} - \sqrt{4}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{4+h} - \sqrt{4}}{h} \cdot \frac{\sqrt{4+h} + \sqrt{4}}{\sqrt{4+h} + \sqrt{4}} \\ &= \lim_{h \rightarrow 0} \frac{4+h-4}{h(\sqrt{4+h} + \sqrt{4})} \\ &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{4+h} + 2)} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{4+h} + 2} \\ &= \frac{1}{\sqrt{4+0} + 2} \\ &= \frac{1}{4} \end{aligned}$$

Point on tangent line: $(3, f(3)) = (3, 2)$.

Equation of tangent line:

$$\begin{aligned} y - f(3) &= \frac{1}{4}(x - 3) \implies y - 2 = \frac{1}{4}x - \frac{3}{4} \\ &\implies y = \frac{1}{4}x + 1\frac{1}{4}. \end{aligned}$$

Problem 3

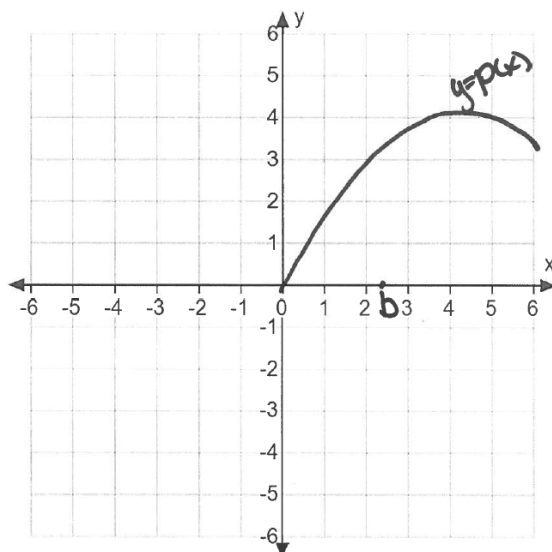


Problem 4

February 21, 2026

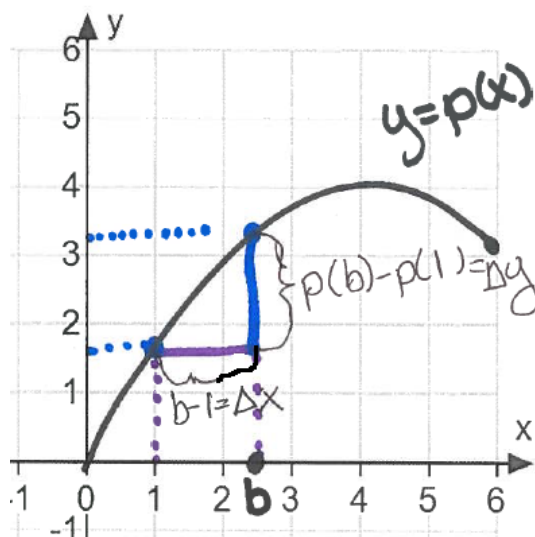
Problem 4

Problem 4 The graph of a function p and a point in its domain, b , are shown in the figure below.



- (a) In the figure above, draw and mark clearly the quantity $\Delta y = p(b) - p(1)$ and the quantity $\Delta x = b - 1$.

Explanation.



- (b) Complete the sentence. The quotient $\frac{p(b) - p(1)}{b - 1}$ is the slope...

Problem 4

Explanation. The quotient $\frac{p(b) - p(1)}{b - 1}$ is the slope of the secant line through the points $(b, p(b))$ and $(1, p(1))$

- (c) Complete the sentence. Provided it exists, the limit $\lim_{x \rightarrow 1} \frac{p(x) - p(1)}{x - 1}$ is the slope...

Explanation. Provided it exists, the limit $\lim_{x \rightarrow 1} \frac{p(x) - p(1)}{x - 1}$ is the slope of the tangent line to the curve $y = p(x)$ at the point $(1, p(1))$

Problem 5

February 21, 2026

Problem 5 An object moving along a straight line has a position given by $s(t) = \frac{1}{t-4}$, where s is measured in meters and t in seconds. Find the velocity of the object at time $t = 6$.

Explanation. $v(6) = s'(6) = \lim_{t \rightarrow 6} \frac{s(t) - s(6)}{t - 6}$. When we check the form of this

$$\begin{aligned} \text{limit, we see that it has form } \frac{0}{0}. \quad s'(6) &= \lim_{t \rightarrow 6} \frac{\frac{1}{t-4} - \frac{1}{2}}{t-6} = \lim_{t \rightarrow 6} \frac{\frac{2 - (t-4)}{2(t-4)}}{t-6} = \\ &= \lim_{t \rightarrow 6} \frac{(2-t+4)}{2(t-4)(t-6)} = \\ &= \lim_{t \rightarrow 6} \frac{(6-t)}{2(t-4)(t-6)} = \lim_{t \rightarrow 6} \frac{-1}{2(t-4)} = \frac{-1}{2(6-4)} = \frac{-1}{4} \text{ m/s.} \end{aligned}$$